Smart Agriculture AI — Comprehensive Farm Report

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1. Soil Analysis

Parameter	Value
Detected Soil Type	Clay
Detection Confidence	52.0%
Detection Method	EfficientNetV2 + SVM

2. Current Soil Conditions (IoT Sensor)

Parameter	Value	Status
Soil Temperature	24.4°C	Normal
Soil Humidity	53.9%	Normal
Soil Moisture	0%	Dry — Needs Irrigation
Irrigation Status	N/A	

3. Weather Conditions

Parameter	Value
City	Bhopal, India
Outside Temperature	25.2°C
Humidity	87%
Rainfall	0.0 mm/hr
Wind Speed	8.3 km/h

4. Field Details

Total Area: **10,000 m²** (2.47 acres)

5. AI Crop Recommendation

Mode: **Multi-Crop (Zone Allocation)**

Crop	Area (ac)	Alloc %	Yield (kg)	Price/kg	Profit (■)	Rating

Wheat	0.71	28.6%	1,071	■21.25	■22,768	Good
Beans	0.71	28.6%	857	■55	■47,143	Good
Cotton	0.65	26.2%	655	■51.4	■33,655	Fair
Rice	0.41	16.7%	750	■15.79	■11,842	Fair
TOTAL	**2.47**	**100%**	**3,333**		**■115,408**	

6. Crop Selection Reasoning

Wheat (Score: 0.6):
• [OK] Compatible with Clay soil
• [OK] Temperature 24.4°C ideal (range: 15-25°C)
• [!] Soil moisture 0% outside ideal (40-70%)
• [i] Current rainfall: 0.0mm/hr
• [i] Soil shows cracking — monitor moisture closely

Beans (Score: 0.6):
• [OK] Compatible with Clay soil
• [OK] Temperature 24.4°C ideal (range: 18-27°C)
• [!] Soil moisture 0% outside ideal (40-70%)
• [i] Current rainfall: 0.0mm/hr
• [i] Soil shows cracking — monitor moisture closely

Cotton (Score: 0.55):
• [OK] Compatible with Clay soil
• [OK] Temperature 24.4°C ideal (range: 21-30°C)
• [!] Soil moisture 0% outside ideal (30-60%)
• [i] Current rainfall: 0.0mm/hr
• [!] Higher risk crop (risk factor: 0.6)
• [i] Soil shows cracking — monitor moisture closely

Rice (Score: 0.35):
• [OK] Compatible with Clay soil
• [OK] Temperature 24.4°C ideal (range: 20-35°C)
• [!] Soil moisture 0% outside ideal (60-100%)
• [i] Current rainfall: 0.0mm/hr
• [!] High water need but low soil moisture
• [!] High water demand — risky in cracked/dry soil conditions

Report generated by Smart Agriculture AI — IoT + Deep Learning System
Cost savings: Farmers typically spend ■10,000-20,000 on lab soil testing.